

Midterm, Math 170S, Winter 2018
Instructor: Elizaveta Rebrova

Printed name: _____
Signed name: _____
Student ID number: _____

Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems.
- Your final answers do not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- Books, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	15	
2	15	
3	10	
4	10	
Total:	50	

1. Do not forget to show all your work.

- (a) (3 points) Let $X_1, X_2, \dots, X_{20}$ are samples taken from normal $N(\mu_X, 2)$ distribution ($2 = \sigma^2$), $Y_1, Y_2, \dots, Y_{40}$ are samples taken from normal $N(\mu_Y, 4)$ distribution, $Z_1, Z_2, \dots, Z_{45}$ are taken from normal $N(\mu_Z, 5)$ distribution.

All X_i 's, Y_i 's and Z_i 's are mutually independent. Additionally, you computed that What is the distribution of a random variable $5\bar{X} - \bar{Y} - 3\bar{Z}$?

For full credit you should either give a name of the distribution and compute its parameters; or compute a density function.

$$\bar{X} \sim N\left(\mu_X, \frac{2}{20}\right)$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ for } X \sim N(\mu, \sigma^2)$$

$$\bar{Y} \sim N\left(\mu_Y, \frac{4}{40}\right)$$

$$\bar{Z} \sim N\left(\mu_Z, \frac{5}{45}\right)$$

$$5\bar{X} - \bar{Y} - 3\bar{Z} \sim N(5\mu_X - \mu_Y - 3\mu_Z,$$

$$\frac{25 \cdot 2}{20} + \frac{1}{10} + \frac{9 \cdot 5}{45}) = N(5\mu_X - \mu_Y - 3\mu_Z, 2.6 + 0.1 + 1)$$

$$\boxed{N(5\mu_X - \mu_Y - 3\mu_Z; 3.6)}$$

Now we additionally computed that

$$X_1 + \dots + X_{20} = 1;$$

$$Y_1 + \dots + Y_{40} = 10;$$

$$Z_1 + \dots + Z_{45} = 3.$$

- (b) (2 points) Give a point estimate for a parameter $5\mu_X - \mu_Y - 3\mu_Z$.

$$\bar{X} = \frac{1}{20}$$

$$5\mu_X - \mu_Y - 3\mu_Z$$

$$\bar{Y} = \frac{1}{4}$$

$$\frac{1}{4} - \frac{1}{4} - \frac{9}{45} = -0.2$$

$$\bar{Z} = \frac{3}{45}$$

You must give numerical answers; but you do not need to simplify them completely, e.g., a sum of several fractions would be completely fine

- (c) (5 points) Find 2-sided 95% confidence interval for a parameter $5\mu_X - \mu_Y - 3\mu_Z$.

$$P\left(-z_{\alpha/2} \leq \frac{5\bar{X} - \bar{Y} - 3\bar{Z} - (5\mu_X - \mu_Y - 3\mu_Z)}{\sqrt{3.6}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

$\alpha = 0.05$

$$5\mu_X - \mu_Y - 3\mu_Z \in [-0.2 - 1.96 \cdot \sqrt{3.6}; -0.2 + 1.96 \cdot \sqrt{3.6}]$$

- (d) (5 points) Find 1-sided 80% confidence interval for the upper bound of the parameter $5\mu_X - \mu_Y - 3\mu_Z$.

$$5\mu_X - \mu_Y - 3\mu_Z \in [\text{undefined} - 0.2 + z_{0.2} \cdot \sqrt{3.6}]$$

$$= [\text{undefined} - \infty; -0.2 + 0.842 \cdot \sqrt{3.6}]$$

2. Let $f(x; \theta) = (1/\theta)x^{(1/\theta)-1}$, $0 < x < 1$, $0 < \theta < \infty$.

(a) (5 points) Find $\hat{\theta}$ = the maximum likelihood estimate of θ .

(see also problem 6.4-8 in the textbook)

$$L(X_1, \dots, X_n; \theta) = \frac{1}{\theta^n} \cdot \prod_{i=1}^n X_i^{\frac{1-\theta}{\theta}}$$

$$\ln L(\theta) = -\ln(\theta) \cdot n + \left(\frac{1}{\theta} - 1\right) \cdot \sum_{i=1}^n \ln X_i \rightarrow \max$$

$$\left(\ln L(\theta)\right)'_{\theta} = -\frac{n}{\theta} - \frac{1}{\theta^2} \left(\sum_{i=1}^n \ln X_i\right) = 0$$

$$\theta n + \sum_{i=1}^n \ln X_i = 0 \quad \left| \theta = - \frac{\sum_{i=1}^n \ln X_i}{n} \right|$$

It gives maximum, since
 $\left(\ln L(\theta)\right)' < 0$ for $\theta > \hat{\theta}$ and
 $\left(\ln L(\theta)\right)' > 0$ for $\theta < \hat{\theta}$.

(b) (5 points) Find $\tilde{\theta}$ = the method of moments estimate of θ .

We will set $\mu_1 = \bar{X} = EX$

So, we need to find EX :

$$EX = \int_0^1 \frac{1}{\theta} x^{\frac{1}{\theta}-1+1} dx = \int_0^1 \frac{1}{\theta} x^{\frac{1}{\theta}} dx = \frac{1}{\theta} \frac{x^{\frac{1}{\theta}+1}}{\frac{1}{\theta}+1} \Big|_0^1$$

$$= \frac{1}{\theta} \cdot \frac{\theta}{\theta+1} = \frac{1}{\theta+1}$$

So, $\bar{X} = \frac{1}{\tilde{\theta}+1}$
 $\tilde{\theta} = \frac{1}{\bar{X}} - 1$

(c) (5 points) (a) Is $\hat{\theta}$ an unbiased estimate of θ ? (b) Is $\tilde{\theta}$ an unbiased estimate of θ ?

For (b), you may use without the proof that $\mathbb{E}(1/\bar{X}) \neq 1/\mathbb{E}(\bar{X})$

$$a) \mathbb{E} \hat{\theta} = - \frac{\sum_{i=1}^n \mathbb{E} \ln X_i}{n}$$

$$\begin{aligned} \mathbb{E} \ln X &= \int_0^1 \frac{1}{\theta} x^{\frac{1}{\theta}-1} \ln x \, dx = \ln x \cdot x^{\frac{1}{\theta}} \Big|_0^1 - \int_0^1 x^{\frac{1}{\theta}} \frac{1}{x} \, dx \\ &= 0 - \int_0^1 x^{\frac{1}{\theta}-1} \, dx = -\theta x^{\frac{1}{\theta}} \Big|_0^1 = -\theta \end{aligned}$$

$$\text{So, } \mathbb{E} \hat{\theta} = - \frac{n(-\theta)}{n} = \theta \Rightarrow \hat{\theta} \text{ is unbiased}$$

$$\begin{aligned} b) \mathbb{E} \tilde{\theta} &= \mathbb{E} \left(\frac{1}{\bar{X}} - 1 \right) = \mathbb{E} \left(\frac{1}{\bar{X}} \right) - 1 \neq \frac{1}{\mathbb{E}(\bar{X})} - 1 = \frac{1}{\mathbb{E} X} - 1 \\ &= \theta + 1 - 1 = \theta \end{aligned}$$

$$\text{So, } \mathbb{E} \tilde{\theta} \neq \theta \Rightarrow \tilde{\theta} \text{ is biased}$$

3. (10 points) Suppose that observations are coming from a distribution with the density function

$$f_X(x) = \theta e^{-\theta x} \quad \text{for } x \geq 0,$$

where θ is an unknown parameter. Additionally you know that θ is either 1, 2 or 3. Prior the experiment, you have no idea which option is more likely (so, let's assume that all options are equally likely). Then you observe in 3 experiments that $X = 3, 2, 5$. Find the MAP estimator of θ . Give numerical value as an answer.

$$P(\theta | X) = \frac{f(X_1, X_2, X_3 | \theta) \cdot P(\theta)}{\sum_{\theta' \in \{1, 2, 3\}} f(X_1, X_2, X_3 | \theta') \cdot P(\theta = \theta')}$$

So, $P(1 | X_1, X_2, X_3) = \frac{\theta^3 \exp(-10\theta)^{\frac{1}{3}}}{111} = \frac{\exp(-10)}{3 \cdot 111}$ — something, a constant, the same

$P(2 | X_1, X_2, X_3) = \frac{\theta^3 \exp(-10\theta)^{\frac{2}{3}}}{111} = \frac{8 \exp(-20)}{3 \cdot 111}$

$P(3 | X_1, X_2, X_3) = \frac{\theta^3 \exp(-10\theta)^{\frac{3}{3}}}{111} = \frac{27 \exp(-30)}{3 \cdot 111}$

To find MAP, we compare red numbers:

- $\frac{1}{3} e^{-10} > \frac{8}{3} e^{-20}$ since $e^{10} > 8$
- $\frac{1}{3} e^{-10} > 9 e^{-30}$, since $27 < e^{20}$

So, $\theta = 1$ gives maximum value of posterior distribution

$$\boxed{\text{MAP } \hat{\theta} = 1}$$

4. (10 points) We have two samples X_1, X_2 taken from the uniform distribution

$$Unif([\theta - 2, \theta + 1]).$$

What is the length of a two-sided confidence interval with confidence level 80% for the estimate $X_{(2)}$ for parameter θ ? $X_{(2)}$ denotes the second order statistic.

$$X_1, X_2 \sim Unif([\theta - 2, \theta + 1]) \quad 1 - \alpha = 0.8$$

So, we need to find c :

$$P\left(\theta \in \left[X_{(2)} - \frac{c}{2}, X_{(2)} + \frac{c}{2}\right]\right) = 0.8$$

$$P\left(X_{(2)} - \frac{c}{2} \leq \theta \leq X_{(2)} + \frac{c}{2}\right) = 0.8$$

$$P\left(\theta - \frac{c}{2} \leq X_{(2)} \leq \theta + \frac{c}{2}\right) = 0.8$$

$$0.8 = P\left(X_{(2)} \leq \theta + \frac{c}{2}\right) - P\left(X_{(2)} < \theta - \frac{c}{2}\right)$$

$$= P\left(X_1 \text{ and } X_2 \leq \theta + \frac{c}{2}\right) - P\left(X_1 \text{ and } X_2 < \theta - \frac{c}{2}\right)$$

$$\stackrel{\text{as } \frac{c}{2} \leq 1}{=} P\left(X \in \left[\theta - 2, \theta + \frac{c}{2}\right]\right)^2 - P\left(X \in \left[\theta - 2, \theta - \frac{c}{2}\right]\right)^2$$

$$= \left(\frac{\frac{c}{2} + 2}{3}\right)^2 - \left(\frac{-\frac{c}{2} + 2}{3}\right)^2 = \frac{\left(\frac{c}{2} + 2 - \frac{c}{2} + 2\right)\left(\frac{c}{2} + 2 + \frac{c}{2} - 2\right)}{9} = \frac{4 \cdot c}{9} = 0.8$$

as $\frac{1}{3}$ is density of $Unif[\theta - 2, \theta + 1]$

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$\frac{c}{2} \leq 1$, so we are ok

$$c = \frac{8}{5} \cdot \frac{9}{4} = \boxed{1.8}$$